Ondansetron

Mechanism of action

- Ondansetron is an antiemetic.
- Ondansetron antagonises (blocks) serotonin receptors centrally in the brain and peripherally in the gastrointestinal tract, resulting in a reduction in nausea and vomiting.

Authorised practice levels

- EMTs: IM.
- Paramedics, ICPs and CCPs: IV and IM.

Indications

- ✓ Clinically significant nausea and/or vomiting.

Contraindications

- ✕ Known severe allergy.
- ✕ Age less than one year.
- ✕ Suspected serotonin syndrome.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ondansetron may be administered if nausea and/or vomiting is severe.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

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Dosage

- 8 mg IV once for an adult. See the paediatric drug dose tables for a child.
- 4 mg IM for an adult, if IV access cannot be obtained. See the paediatric drug dose tables for a child.
- The IM dose may be repeated once after 20 minutes if required.
- One IV dose may be administered in addition to one IM dose, if clinically significant nausea and/or vomiting persists and IV access is subsequently obtained. For example, in this setting a maximum total dose of 12 mg may be administered to an adult.

Administration

- The preferred route of administration is IV.
- Administer IV undiluted.
- Administer IM undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Headache.
- Flushing.

Usual onset of effect

- IV: 2-5 minutes.
- IM: 5-10 minutes.

Usual duration of effect

- 4-8 hours.

Usual preparation

- Ampoule containing 4 mg in 2 ml.

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- Ampoule containing 8 mg in 4 ml.

Pharmacokinetics

- Ondansetron is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Ondansetron has been reported to prolong the QT interval, particularly if high doses are administered in conjunction with other medicines that also prolong the QT interval, for example erythromycin. However, one or two doses in this setting is safe.

Additional information

- Prophylactic administration of ondansetron is not routinely required. Consider administering ondansetron if the nature of the patient's injuries and transport position are such that vomiting would be particularly problematic.
- Ondansetron should not be administered for vomiting associated with an altered level of consciousness because it is rarely effective in this setting.
- Ondansetron has been reported to further prolong the QT interval in patients known to have a prolonged QT syndrome. This generally involved repeated and/or high doses and one or two doses is safe, even if the patient is known to have a prolonged QT syndrome.
- Nausea and/or vomiting in the context of serotonin syndrome must not be treated with ondansetron as this may worsen serotonin syndrome.

More Information

Version history

11 September 2023 1.0.4

Update: Additional information updated to include serotonin syndrome.